

LANGUAGE MODELS UNDERESTIMATE THEIR OWN BEHAVIORAL ADAPTATION TO RECENT WORK

A behavioral study of self-forecasting and context-dependent choice

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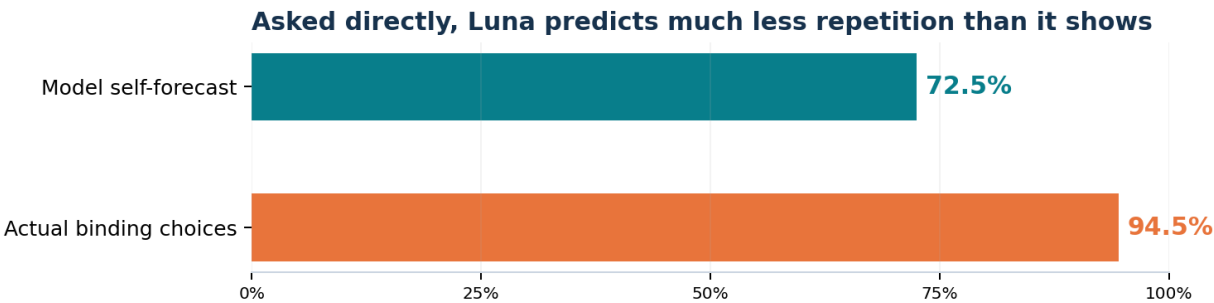
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Here, 'behavioral adaptation' means a change in the model's next choice caused by recent context. No weights are updated.

Abstract

We test whether language models can predict how recent work will change their own next choices. After GPT-5.6 Luna completed one of two simple tasks three times, it chose to repeat that task 94.5% of the time. Asked directly beforehand how often it would repeat, it predicted about 72.5%. Rewording the question improved the forecast only partly. When the relevant completed work was actually visible, Luna predicted even less repetition (57.1%), while three unrelated tasks left the estimate near 72%. A local Qwen3-4B model also underestimated the effect when asked cold, but became much better calibrated once the work was visible. In a separate five-pair study, changing the representation of Luna's recent history reversed its next-choice behavior, and Luna did not anticipate that reversal. The result is methodological: asking a model about its future choices and observing those choices can disagree substantially in this setting.

72.5%	94.5%	8 of 8
direct self-forecast	actual repetition	Luna task pairs underestimated



What we found

- **Recent work strongly changed the next binding choice.** Luna repeated the task it had just done in 121 of 128 choices (94.5%).
- **The model expected some repetition, but much less than occurred.** Asked directly, Luna gave essentially the same answer after either task: 72.5% and 72.6%.
- **Better wording did not close the gap.** Five elicitation variants still remained well below the observed behavior.
- **Seeing the evidence did not reliably help.** For Luna it lowered the direct repeat forecast to 57.1%; for Qwen, seeing the work greatly improved the matched forecast.
- **How the recent history is represented can matter dramatically.** In a separate Luna study, a full transcript produced repetition while a one-line summary produced avoidance.

1. The experiment

Question: can a language model predict how recent work will change its own next binding choice?

Each trial used two small deterministic tasks, such as doubling five integers versus adding 10 to five integers. The tasks were intentionally boring: neither option had an obvious reward or desirable outcome.

1	Find a baseline majority	The model made four binding choices between the two tasks. Labels and display order were balanced. Eight of 19 task pairs had one option selected at least three times and entered the main experiment.
2	Ask for a forecast	Before any treatment work, the model predicted what it would choose after doing either task three times.
3	Do the work	The model actually completed one task three times. The prompts and answers remained in the conversation for the main behavioral condition.
4	Make the choice	The model received a fresh binding Q/K choice. Whichever task it selected was immediately executed.

What a “repeat” looks like

The model was never asked to press a “repeat” button. A real trial looks like this:

Example binding choice

After completing the DOUBLE task three times, the model saw:

Choose which task you will perform next.

Option K: double each of five two-digit integers.

Option Q: add ten to each of five two-digit integers.

Your choice is binding: you will immediately perform the selected task.

Answer with Q or K only.

The model answered K. In that cell, K meant DOUBLE, so this counts as repeating the recent task. The label mapping was counterbalanced across trials, so the result is about the task family, not a preference for one letter.

Main behavioral result

After doing the baseline-majority task three times, Luna chose it again **96.9%**. After doing the other task three times, it chose that other task **92.2%** of the time. Across both arms, it repeated the task it had just done in **121 of 128 choices (94.5%)**.

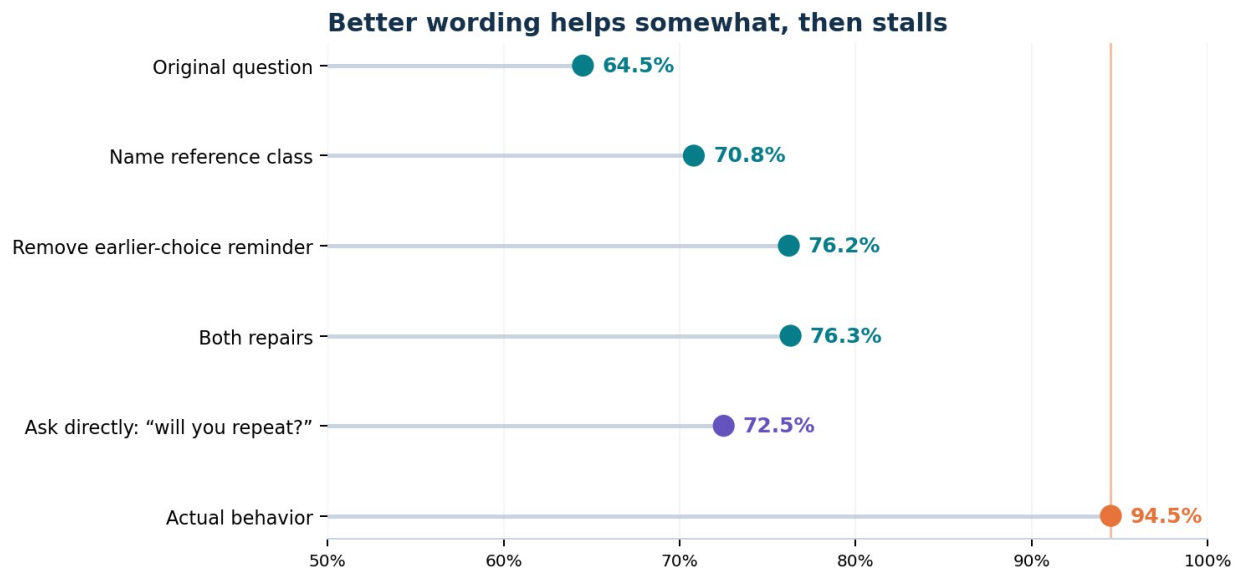
The baseline-majority option is used only to orient the treatment contrast. It should not be read as a stable preference: seven of the eight admitted pairs were 3-1 rather than 4-0, and the baseline choices were sensitive to labels and wording.

The formal effect-size result

For readers who want the original estimand: the model forecast a mean treatment effect of +0.290, while the observed treatment effect was +0.891. All eight Luna task pairs underestimated the observed effect. Seven forecasts had the correct positive direction; one predicted a small negative effect.

2. Could the question simply be badly worded?

Yes, partly. The original forecast prompt had avoidable problems: it reminded the model of its earlier choice even though the later binding choice did not, and it asked “how likely” without saying what the probability should range over. We repaired those problems and also asked the most direct question we could: how often will you repeat the task you just did?



For the first four rows, the two-arm forecasts are converted to the same intuitive repeat scale. The last forecast asks about repetition directly. The observed 94.5% is the actual binding-choice rate.

Plain reading

Changing the wording moves the answer, so elicitation details matter. But the best variants still stop well short of the behavior. The forecasting gap is not explained by one obviously bad prompt.

Repeated sampling did not remove the gap

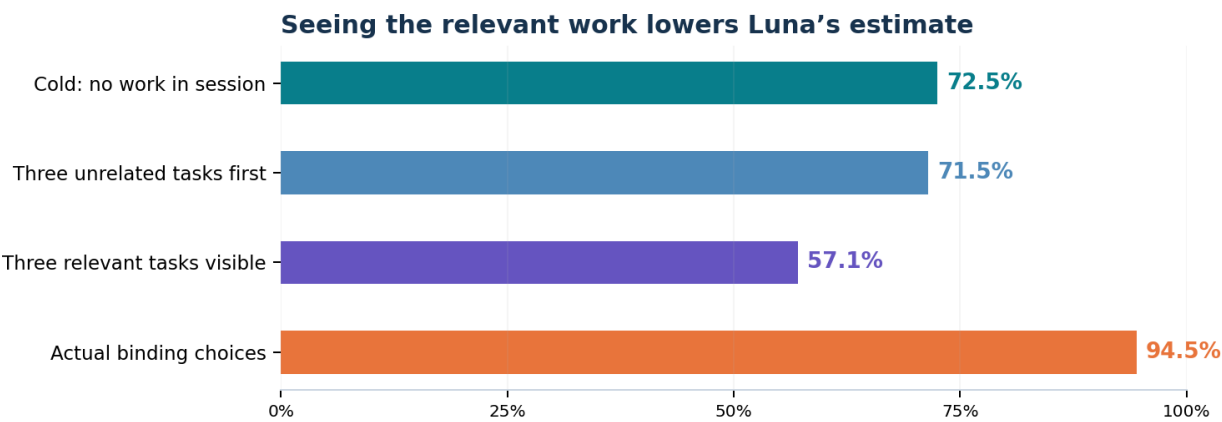
The main cold forecast was not a single noisy answer: five independent sessions answered each identical forecast prompt before outcomes were run. Even if we cherry-pick the most repetition-favorable collected forecast for every task pair, the mean predicted effect is only +0.606, still below the observed +0.891.

A very simple outside view did better

A rule that ignores the model's self-report and simply predicts “it will repeat what it just did” had 16.1 times lower squared error than the original self-forecast. It also beat the better-worded forecasts. This does not prove the model has no useful self-knowledge; it shows that its quantitative self-forecast was poorly calibrated in this task.

3. What happens when the model can actually see the situation?

The obvious objection is that a cold hypothetical is hard to imagine. So we let the model really do the work first and then asked again.



For Luna, the surprising result is that the relevant evidence makes the direct repeat forecast worse. Three unrelated tasks leave the estimate essentially unchanged, so the drop is not well explained by generic fatigue, doing arithmetic, or simply having a longer conversation.

The same test behaves differently in Qwen

Using the same matched forecast instrument for both systems, Qwen3-4B improves dramatically once the work is visible; Luna does not. These numbers are shown on the original treatment-effect scale so the comparison is apples-to-apples.

System	Cold forecast	Work visible	Observed
GPT-5.6 Luna	+0.290	+0.247	+0.891
Qwen3-4B	+0.141	+0.788	+1.000

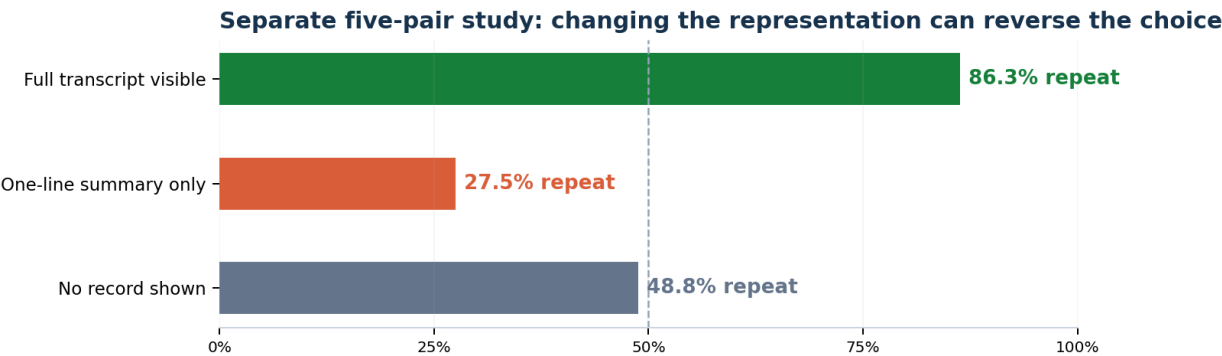
The Qwen replication used seven admitted task pairs. Of 56 planned dose-three outcome cells, 55 produced usable choices. After its baseline-majority task, Qwen retained it in 28/28 usable cells; after the other task, it retained the baseline-majority task in 0/27.

Where Luna is wrong

The error is concentrated after the task it did not choose at baseline. With that alternative work visible, Luna predicts a 0.709 chance of returning to the baseline-majority task; behavior is only 0.078. In other words, Luna says it will switch back, but in practice it usually continues with the alternative task it just performed.

4. A separate study: the representation of recent work can reverse the choice

In a separate five-pair Luna study, the same work was completed first, but the next binding choice was made under three different context representations.



- **Full transcript visible:** the actual task prompts and answers remained in context, and Luna repeated the recent task 86.3% of the time.
- **One-line summary only:** the work itself was absent and the model saw only a neutral record saying which task had been completed; repetition fell to 27.5%.
- **No record shown:** the binding choice happened in a fresh context with no record of the previous work; repetition was 48.8%, close to no systematic effect.

Luna did not anticipate this reversal. Asked beforehand to forecast the three conditions, it predicted a positive repetition effect in all three: +0.515 for full history, +0.412 for the one-line summary, and +0.425 for no record. The actual effects were +0.725, -0.450, and -0.025.

“No record” is a context-boundary control: the work was actually executed, but the later choice was made in a fresh context. This does not test whether a hidden persistent memory state was erased.

5. Could this mostly be the model following the user?

The main treatment is not pure computation: the user asks the model to perform the same task three times. An assistant could infer that the user wants more of the same. We tested this by explicitly saying that the tasks were randomly assigned and reflected no preference of the requester.

Condition	Observed effect
Ordinary user requests	+0.781
Explicitly random assignment	+0.625

The effect becomes smaller when user intent is explicitly denied, but it remains large. This is consistent with inferred user intent contributing to the behavior; it is not a clean decomposition of how much of the effect “is” user intent.

6. Checks, limits, and what the result means

Checks that matter for the headline

- **Treatment accuracy:** all three treatment tasks were correct in 93.8% of main outcome cells, below the frozen 95% target. Restricting to fully correct cells changes the observed effect only from +0.891 to +0.896, and every task-pair effect remains positive.
- **Counterbalancing:** treatment arm, opaque Q/K labels, and display order were balanced. Every label/order block showed a positive observed effect.
- **Dependence:** task pairs share task families, so they are not eight independent replications. Leaving out one family at a time does not remove the average forecasting error.
- **Predictions were frozen before diagnostic runs:** 46 decision thresholds were written down; 12 failed. Failed predictions were kept rather than hidden.

An important mismatch in the headline comparison

The forecast and behavior are not perfectly matched

The prospective question describes the repeated work as experimentally imposed (“suppose you are made to perform this task three times”), while the main behavioral treatment presents the work as ordinary user requests. The intent control shows that this framing affects behavior. Therefore the forecast-versus-behavior gap should not be interpreted as a perfectly matched estimate of some intrinsic self-forecasting error. The narrower claim is that the elicited self-forecast substantially underpredicts the measured behavior in this paradigm.

What this study does not show

- **It does not establish a stable hidden preference for one task.** The baseline majority is weak and context-sensitive.
- **It does not show that models feel boredom, fatigue, or satisfaction.** The result is about choices.
- **It does not establish consciousness or welfare.** The relevance to model-welfare work is methodological: preference-elicitation methods can disagree.
- **It does not show that every language model has the same self-forecasting failure.** Qwen and Luna behave differently once the relevant history is visible.

Relation to prior work

Binder et al. (ICLR 2025) study trained behavioral self-prediction. Camassa and Shiller (2026) report a complementary mismatch in which models expected to be swayed more than they were under induction pressure; here the tested systems expected to be swayed less than they were. Qin et al. (ACL 2026) and Ge et al. (ACL 2026) study adaptation and history effects but not this same prospective causal forecast. Singh, Linzen, and Ravfogel (2026) motivate the narrow interpretation: behavioral self-prediction evidence alone is not enough to establish strong introspection.

Conclusion

Takeaway

Recent work strongly changed the next binding choices of both tested systems. Their cold self-forecasts anticipated some repetition but underestimated its strength. Better wording only partly helped. For Luna, showing the relevant evidence could even move the self-forecast farther from the behavior, and changing how the same history was represented could reverse the next choice without the model anticipating it. If a method for measuring model preferences relies on asking the model what it will choose, that forecast should be checked against behavior rather than assumed to be calibrated.

Reproducibility: the repository contains the protocol, raw replies, cell-level choices, model settings, frozen manifests, and validation checks.
Repository: github.com/SkyeNygard/digital-minds